Notes: 1st official meeting of 7000-X Committee

1st Meeting: concerning 7090 & its improvements Nov 10, '58

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7090 successor?

(Jones) if Transac S-4000 comes up — w

7097X compatibility: how compatible is it? "almost entirely"

no copy add & carry
copy skip
load Drum address
} still decoded treated as no-ops.
709 "compatibility switch" copyz trap

compatibly { (1) an interpretive prog.
(2) some hardware. makes drum simulation possible

possible improved version

Memory may go to 2.0 μs

presently using 200 mps 5 mc
are designed to use now. 167 mps 6 mc

mem cycle = 12 machine cycles adds can start EK-0 or EK-6

7090 adder can be speeded up to 800 mps at least

(presently do 300 mps)
to go thru adder, — but takes 1 μs total

can skip 1's as well as 0's get twice as fast as now

(Jones' Table) Fixed Mpy: ER-0, ER-6

	Present	2	3	4	6	12
skip 0 only	37.3 μs	34.4 μs	34.9	29.8	29.1	26.8
skip 0 + 1		28.4	28.2	25.4	24.5	23.9

(2) can shift 2 bits 01 10 in addition

Ex: can shift only 1 bit in addition above using random array of bits

(3) can start ER0, ER4, ER8 (slightly worse — takes whole cycle to shift 1 out)

(12) full asynchronous

Floating add: presently 7 cycles 10 preshifts, 4 postshifts (can do is 7.1 cycles)

3 cycles 4 preshifts, 1 postshifts (3.5 cycles)

(1 cycle = 2.4 μsec)

So a factor of 2 in FL add with present circuits

→ can use more circuits

Transfers, etc could be be dropped to 1 cycle.

Time charts of 7090 (see other sheet.) question of memory read out in 1.4 μsec

question 7 cycle × 0.2 = 1.4 μs data is available??

or

question: Mem. reliability?

noise ?

IKMPD: a big standard machine 48 bit being rumored ←

also LARC-2: about 20 to 25 times 704 speed. ←

question of 705 III-TX? considered announced?

7090: Program compatibility with 709 - still important ground rule,

summary:

-- So can go about ~2 betts than present 7090

So result ~~8 than~~ 8 times 709 present circuitry,
10 times " with betta circuitry,

II. - Also consider improved I/O area.

STRETCH 1. { High speed disk file
(or) large capacity drum } can't connect thru regular DSU

SEPD STRETCH 2. Serial disk file (binary addressing) -- need a "locate instr."

3. SWIFT Tapes.

4. Console type serial reader-printer, (request is for printer now)

5. Stromberg Carlson printer (online)

6. Chain printer

7. High speed card reader. 2000 cpm

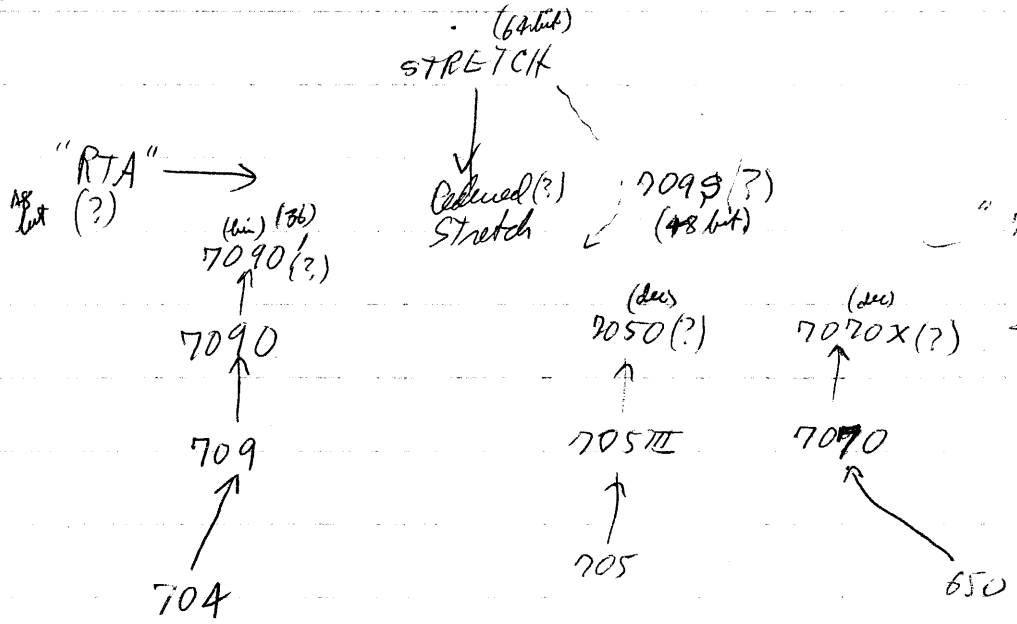
8. High speed punch.

SEPD & DP-PE-SE 9. "real-time package" - 36 lines to give & receive signals from channel - DSU. External signal interrupt

10. paper tape?

Question: I/O in general - Tapes to peripheral - ← See C. Poland

arena into which machine must go;



(1) dec ↔ binary.
(2) word size
(3) prog. compatible.

Goals — Things to consider - shortcomings - etc.

"Holy grail" (scientific - Commercial)

Exchange a ?

Scratch-pad mem. ? (eg. 709S)

IDP - interrupt system.

Sych - asynch circuits.

decimal - binary

checking - how much?

- control word philosophy,
- logical ability.
- Variable field
- multiprogramming
- indexing - no. & flexibility.

(Cochran - Hopper ref. core switching circuits) ← Ref

"Bill Cadden" - from Bell Lab ← ref

other areas:

question:
- what about commercial?
- SABRE
- Service bureau operation.
- multiprograming - dump machine

(IDP) relative addressing
multiple CPU's - machine-to-machine communication.
- priority control.
at high speed.

Pricing - very hard to figure - discontinuous

→ fixed word length is very hard on IDP-type probs. (36 bits is short)

interrupt system: (ref 7070) ←

- one on each channel.

Competition:

Univac LMR ?

RTA

Transac S-4000

LARC-2

(?) IBM-310 - multi machine (Jim McDonald / Morgan) ref ← Endicott

IDP processor - 2 or more machines & machine to file

better than any other IDP now.

Bell Tel machine (Bill English / Jerry Walters) ← ref

Transac to United Aircraft - (Jerry O'Hara) ←

Fortran ?

Weather: starts with 7 stations - grows, 5000 nd/mi

RCA - Sperry Rand - Stanley Carlson

IBM - Western Elec. - ?

Philco

United Aircraft - Travelers weather - (709 TX)

IT&T

Cook Electronics